

Assignment 1

Introduction to Computing For the Social Sciences

WS 2025/26 — **Deadline: 09.11.2025, 23:59**

If you have any questions regarding the assignment or need guidance on how to approach these tasks, please make sure to visit the tutorial sessions.

The tutorials introduce the techniques and packages needed as well as providing guidance and materials to structure the approach to solve these tasks.

Task 01 - GitHub Setup

Create a GitHub account (if you do not already have one) and enter the GitHub classroom (you should have received an email). From this point onward all the materials are going to be provided over GitHub and we also expect you to upload your assignment solutions there.

You can find detailed instructions on how to set up everything in the materials folder.

A general overview of what to do is provided below:

- ☐ Create SSH key pair
- ☐ Fork ICSS repository
- ☐ Pull the 2nd part of the assignment from GitHub
- ☐ Create and push a text file with your name in to your personal submission folder (to test the GitHub workflow)
- ☐ Create the required directory structure

Task 02 - Pandas DataFrames and Data Visualization

Use the csv-files provided ([world-population-by-country-2020.csv](#) & [countries-continents-capitals.csv](#)) to create two visualizations:

1. A bar chart displaying the absolute population per continent in descending order
2. A violin plot displaying the distribution of the continents populations

Outline the steps you have taken to create the visualisations and write a short interpretation of the violin plot.

Task 03 - Lambda Functions and List/Dictionary Comprehension

1. Given the dictionary `people` below, write a program that returns the person closest to their 50th birthday (if there are multiple people with the same distance, it is enough to return one). **Use a lambda function for this task.**
What are the benefits and problems of using lambda functions? Give a short answer.

```
people = [{'name': 'Laurie Pineda', 'age': 39}, {'name': 'Tommy Harper', 'age': 21}, {'name': 'Jeffery Short', 'age': 54}]
```

2. Use list comprehensions to write a Python function `remove_long_words()`. The function should match the description in the docstring. **Use at least one list comprehension in this task.**

```
def remove_long_words():
    """
    For a given string and number, this functions outputs all the words
    in the string that are smaller than or equal the given integer

    Input:
    -----
    - sentence: string
        A string containing words that are separated by whitespaces

    - max_length: integer
        The maximal length of the words to keep

    Return:
    -----
    - result: list
        A list of all words in the sentence whose length is less
    than or equal the given integer
    """
    ...
```

Task 04 - Social Science Application - Senate Election Prediction

We have provided you with two datasets. One containing the predictions for the US Senate 2022 Midterm Election from fivethirtyeight.com and one containing the (preliminary) US Senate Midterm Election results from “cnn.com”. Both datasets are stored as a file in JSON format. The goal of this task is to determine the accuracy of the election prediction. (accuracy: correct cases/all cases).

Hint: Use list or dictionary comprehension.